

QT PDF Editor

Sprint 2 – Initial Project Report

Repository: quantiradev/qt-pdf-editor

Branch: sprint2

Version: 0.1 (Initial Planning)

Classification: Internal Engineering Documentation

1. Executive Summary

Following the successful completion of Sprint 1, Sprint 2 focuses on transforming QT PDF Editor into a complete document management and productivity platform. This sprint introduces cloud-based data persistence using MongoDB Atlas, advanced editing tools, document conversion, PDF security, compression, image scanning, and document comparison capabilities.

2. Sprint Objectives

- Integrate MongoDB Atlas for persistent cloud storage.
- Build a professional toolbar similar to Microsoft Word and Google Docs.
- Add PDF ↔ Word conversion.
- Implement password protection and password removal.
- Compress PDF documents.
- Convert images into PDF.
- Compare two PDF documents.
- Improve backend scalability and maintainability.

3. Sprint Scope

3.1 Database Integration (MongoDB Atlas)

- User Management
- File Metadata
- Document History

- Activity Logs
- User Preferences

3.2 Professional Toolbar

- File, Edit, View, Insert, Annotate, Protect, Convert, Help
- Undo/Redo, Zoom, Select, Hand Tool
- Text, Shapes, Highlight, Drawing, Signature, Watermark

3.3 PDF Conversion

- PDF to Word
- Word to PDF

3.4 PDF Protection

- Lock PDF with Password
- Unlock Protected PDF

3.5 PDF Compression

- Low, Medium and High Compression
- Quality Preservation

3.6 Scan Images to PDF

- Multiple Image Upload
- Page Ordering
- Rotation
- PDF Generation

3.7 Compare Two Documents

- Side-by-side comparison
- Highlight Added, Deleted and Modified Content
- Page-by-page Difference Detection
- Difference Summary

4. Proposed Architecture

Next.js Frontend

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FastAPI Backend

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└─ MongoDB Atlas

└─ PDF Processing Engine (PyMuPDF)

└─ Conversion Engine

└─ Comparison Engine

└─ File Storage

5. Technology Stack

Frontend: Next.js + React + TypeScript

Backend: FastAPI

Database: MongoDB Atlas

Authentication: JWT

PDF Processing: PyMuPDF

Conversion: LibreOffice / python-docx

Compression: PyMuPDF

Image Processing: Pillow / OpenCV

Comparison: PyMuPDF + Text Comparison Engine

6. Expected Deliverables

- MongoDB Atlas Integration
- Professional Toolbar
- PDF ↔ Word Conversion
- Password Protection Module
- PDF Compression Module
- Image to PDF Module
- Compare Documents Module
- Enhanced APIs
- Improved UI/UX

7. Risks & Challenges

- Maintaining formatting during PDF conversion
- Efficient compression without quality loss
- Secure password encryption
- Large document handling
- Accurate document comparison
- Database migration and optimization

8. Success Criteria

Sprint 2 will be considered successful if all planned modules are integrated successfully, MongoDB Atlas stores application data reliably, document comparison identifies differences accurately, and all Sprint 1 features remain stable.

9. Future Roadmap (Sprint 3)

- OCR
- AI Document Chat
- AI Summarization
- Digital Signatures

- Real-time Collaboration
- Cloud Sharing
- Version History
- Role-based Access Control